EFY TESTED Ashwini K. Sinha

A HEAT-SENSING SWITCH That Trips Before Things Burn

Smoke alarms can't tell burning wire from incense, but this thermal-imaging switch sees heat itself and cuts power automatically. Within just '6 hours', you can build this 'medium' complexity device for under ₹4400!

ASHWINI KUMAR SINHA

Smoke alarms can't tell burning wire from incense. This thermal-imaging smart switch sees heat itself, so cooking fumes and incense won't fool it. Built around the ESP8266-12F Wi-Fi module and an AMG8833 thermal grid sensor, it watches for overheating wires, potential short circuits, excessive current-related heating in power lines, elevated boiler surface temperatures, and hot SMPS units or PCBs within the sensor's 0-80°C measuring range. When the measured temperature exceeds a preset threshold, it automatically switches off the connected AC load. It also supports manual control and IoT-based operation through a wireless web dashboard, which streams live thermal images and temperature readings and lets you configure the threshold. An opto-TRIAC (VO2223A) switches AC bulbs and appliances with loads below 1A.

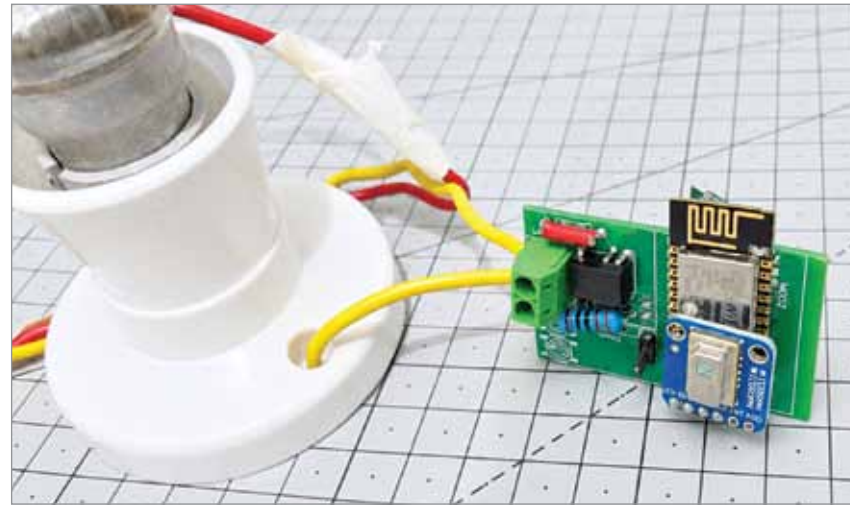


Fig. 1: Author's prototype

BILL OF MATERIALS

Component	Value/Type	Quantity	Cost (₹)
ESP-12F Wi-Fi module	-	1	149
Thermal sensor	AMG8833	1	3899
Opto-TRIAC	VO2223A-X001	1	225
Voltage regulator	AMS1117-3.3, 3.3V 1A	1	5
Resistor	120Ω, 1/4 W, 5% carbon film	1	Miscellaneous ~122
Ceramic capacitor	22μF	1	
Ceramic capacitor	10μF	1	
Tactile switch	TSC005A2518A	1	
2-pin connector	-	1	
Screw terminal	-	1	
PCB	-	1	
Estimated total			**4328**

READY TO BUILD IT YOURSELF?

Head to **https://efy.co.in/gigh53** for the complete circuit diagram, component list, step-by-step build instructions, troubleshooting tips and demo videos. Everything you need to take this project from idea to working prototype is waiting for you there!

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